

CVPR 2023 | A Pose Estimator That Tells You How Wrong It Might Be

NVIDIA Research adds conformal keypoint detection and geometric uncertainty propagation to the standard two-stage pipeline, producing the first object pose estimate with a provable, computable worst-case error bound.

Estimating an object's 6D pose (its three-dimensional rotation and translation relative to the camera) is a workhorse of augmented reality, autonomous driving, robotic manipulation, and space robotics. The dominant recipe is a two-stage pipeline: a neural network first detects semantic keypoints on the image, then a geometric solver recovers the pose by minimizing reprojection error through Perspective-n-Points (PnP). This pipeline tops standard benchmarks, but it hands back a single pose with no statement about how much to trust it.

For a demo that is a minor inconvenience. For a self-driving car or a robot arm it is a real problem. The pipeline gives no guarantee that the detected keypoints are close to the truth, no guarantee that the optimizer found the right pose rather than a plausible wrong one, and no formal bound on the worst-case error between the estimate and the groundtruth. A system that must act on a pose would like to know not just the estimate, but how far off it could possibly be.

This paper, from Heng Yang and Marco Pavone at NVIDIA Research and presented at CVPR 2023, closes that gap. It keeps the familiar two-stage structure and injects two ingredients: conformal keypoint detection, which turns heuristic heatmap detections into prediction sets that provably cover the true keypoints with a user-chosen probability, and geometric uncertainty propagation, which pushes those sets through the geometry to a set of poses that covers the true pose with the same probability. The result is, to the authors' knowledge, the first object pose estimator that attaches a provable and computable worst-case error bound to its output.

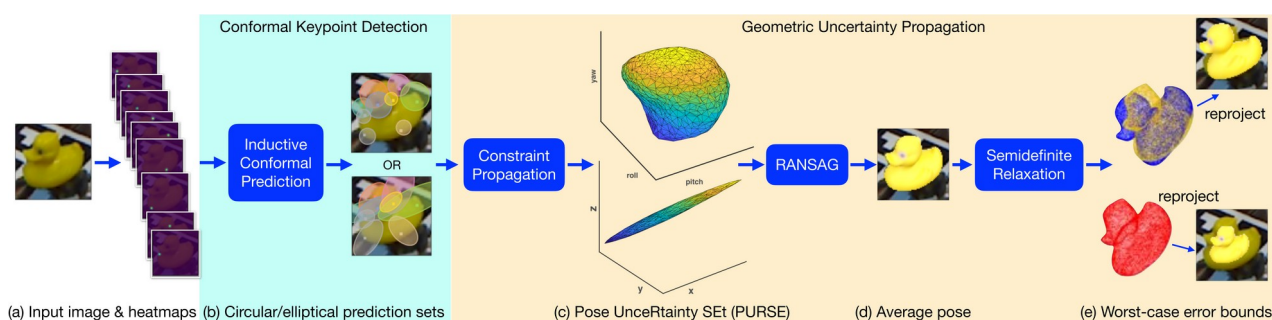


Figure 1. The full pipeline. From (a) an input image and its keypoint heatmaps, conformal prediction produces (b) circular or elliptical prediction sets that cover the true keypoints with a chosen probability (e.g. 90%). Propagating these sets through the pose geometry yields (c) the Pose Uncertainty SEt (PURSE), which contains the true pose with the same probability. RANSAG samples PURSE to produce (d) an average pose, and a semidefinite (SDP) relaxation computes (e) the worst-case rotation and translation error bounds (blue duck: worst rotation; red duck: worst translation).

Paper: <https://arxiv.org/abs/2303.12246>
<https://github.com/NVlabs/ConformalKeypoint>

Code:

The gap: accurate on average, silent about the worst case

The authors name three challenges that stand between a two-stage pipeline and a trustworthy pose. First, keypoints from a neural network can be arbitrarily wrong: an occluded or ambiguous part produces an outlier detection far from the true location. Second, the geometric stage must reject those outliers, which is a nonconvex optimization; fast heuristics like RANSAC are the standard tool but cannot certify that they found the global optimum, and they can fail silently. Third, and most fundamentally, none of this delivers provably correct uncertainty quantification: there is no formal, probabilistic bound on the error between the estimate and the groundtruth. Recent work could certify that the optimization reached its global optimum, which helps with the second challenge, but a certified optimizer still does not tell you how close the resulting pose is to the truth.

Conformal keypoint detection: from heatmaps to prediction sets

The first ingredient replaces a point detection with a set that comes with a coverage guarantee. The tool is inductive conformal prediction. The authors set aside a calibration set of labeled images and, for each one, compute a nonconformity score that measures how surprised the detector is by the true keypoint. Sorting these scores gives a threshold: any new detection is expanded into the set of pixels whose score falls under that threshold. The guarantee is clean and distribution-free: for a target error rate ϵ , the true keypoint lies inside its prediction set with probability at least $1 - \epsilon$ (90% at $\epsilon = 0.1$), as long as calibration and test images are exchangeable.

The shape of the set depends on how the score is defined. A peak design measures the scaled distance between the true keypoint and the heatmap's single most probable pixel, producing a circular set (a ball). A covariance design measures a Mahalanobis distance over the top detections, producing an elliptical set that can stretch along the direction the detector is least sure about. Circular sets are simpler; elliptical sets adapt to anisotropic uncertainty and, as the experiments show, usually cover at least as well.

Geometric uncertainty propagation: from keypoint sets to a set of poses

A guarantee on keypoints is only useful if it survives the trip to the pose. Each keypoint prediction set becomes a constraint on the 6D pose: the pose must reproject the three-dimensional model point into that keypoint's set. Intersecting all these constraints defines the Pose UnceRtainty SEt, or PURSE, the set of all poses consistent with every keypoint's prediction set. Because each keypoint is covered with probability $1 - \epsilon$, PURSE contains the true pose with the same probability. The catch is that PURSE is a nonconvex subset of $SE(3)$, so you cannot simply read a point estimate or an error bar off it.

The paper extracts two things from PURSE. To get a usable point estimate, RANDOM SAMple averaGing (RANSAG) repeatedly samples three keypoints, solves the minimal P3P problem, keeps the poses that fall inside PURSE, and averages the accepted rotations and translations into an average pose. To get an error bound, the authors ask a worst-case question: over all poses still inside PURSE, how far can one get from the average pose? That

maximization is nonconvex, but it admits a semidefinite (SDP) relaxation whose solution is either exact or a valid upper bound. Whenever PURSE actually contains the true pose (which happens with probability 1 minus epsilon), that number is a certified worst-case bound on the rotation and translation error.

Does the guarantee hold in practice?

The experiments run on LineMOD Occlusion (LM-O), a challenging benchmark of 1214 test images showing 8 objects on a table under heavy occlusion. The 200 images selected by the BOP challenge serve as the calibration set that conformal prediction needs, and all 1214 are used for testing. The authors evaluate four setups that cross two choices: keypoints from groundtruth bounding boxes or from a Faster-RCNN detector, and prediction sets that are balls or ellipses.

The first thing to check is coverage: does a target of 90% actually deliver 90%? Conformal prediction guarantees this in expectation over calibration sets, but any single calibration set gives a slightly random coverage. Figure 2 shows how that randomness shrinks as the calibration set grows.

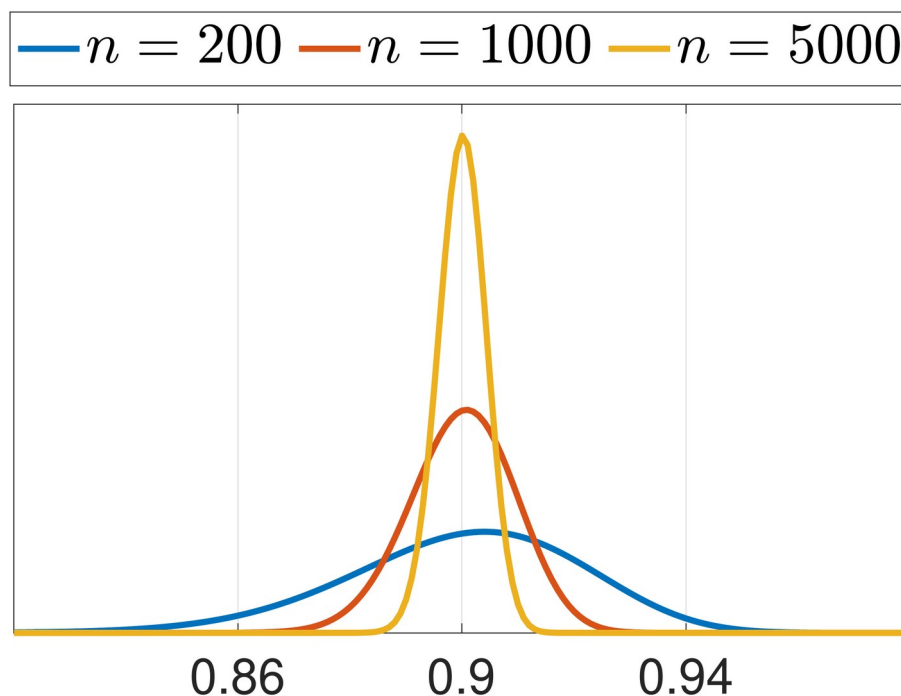


Figure 2. Distribution of the achieved coverage at a 90% target (epsilon = 0.1) for different calibration-set sizes n . Even at $n = 200$ the distribution centers on 0.9; as n grows to 1000 and 5000 it concentrates tightly around the target, so a larger calibration set makes the realized coverage more reliable.

The takeaway is that even the modest 200-image calibration set available on LM-O is enough to center the coverage on the target, and more calibration data mainly tightens the spread.

Coverage also has to hold object by object, not just on average, otherwise a strong overall number could be hiding a failure on one hard category.

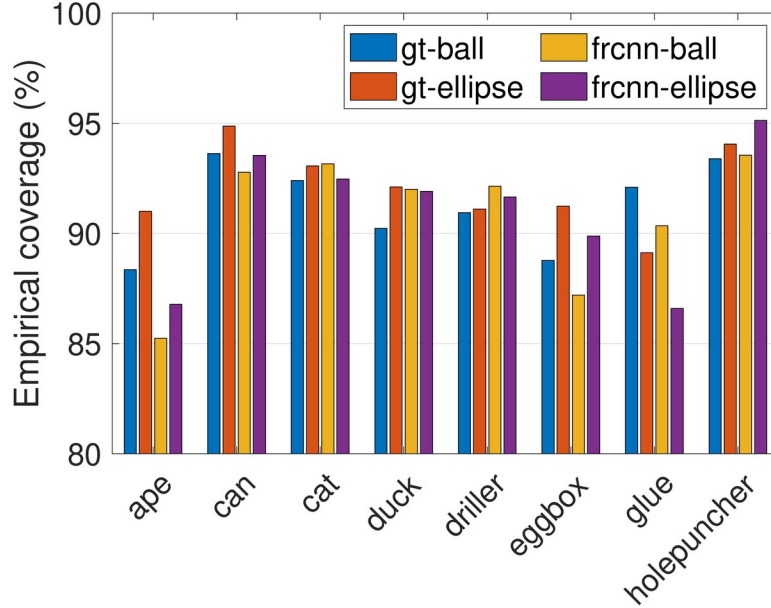


Figure 3. Empirical coverage per LM-O object under the four setups (gt-ball, gt-ellipse, frcnn-ball, frcnn-ellipse). Across all 8 objects the conformalized prediction sets sit at or above the 90% target, confirming valid coverage in practice.

Every object clears the 90% line under every setup, so the coverage guarantee is not an average that hides failures on hard objects. The bounds themselves behave as promised too: in the paper's scatter plots of certified bound against actual error, every case where PURSE covers the true pose stays below the diagonal, meaning the bound is never smaller than the real error when the guarantee holds.

Accuracy: the guarantee is not paid for with worse poses

A guarantee would be a poor trade if it wrecked accuracy. It does not. Judged by the 2D-projection metric (a pose counts as correct if its mean reprojection error is under 5 pixels), the average pose from PURSE matches or beats strong baselines. With groundtruth bounding boxes and elliptical sets it reaches 70.6% mean success, ahead of PVNet, the strongest baseline, at 61.1%. With Faster-RCNN detections the numbers are close to PVNet rather than ahead, which the authors report plainly.

Table 1. Mean 2D-projection success rate on LM-O (a pose is correct if the mean reprojection error is below 5 pixels). Our average pose is shown at epsilon = 0.1.

Method	Success rate (%)
Tekin	6.16
PoseCNN	17.20
Oberweger	60.90
PVNet (strongest baseline)	61.06
Ours: gt-ball	67.33
Ours: gt-ellipse	70.61
Ours: frcnn-ball	56.93
Ours: frcnn-ellipse	57.73

Two knobs shape this trade-off. The error rate epsilon acts as a dial: a smaller epsilon (0.1) gives larger, more conservative prediction sets and stronger coverage, while a larger epsilon (0.4) shrinks the sets, raises the average pose's success rate, and lowers coverage. And elliptical sets, built from the covariance design, generally match or beat circular ones, which is why gt-ellipse leads the table.

What it means, and where it stops

The lasting message is that a standard two-stage pose estimator can be upgraded into one that reports how wrong it might be, with a genuine statistical guarantee, and at little or no cost in accuracy. Conformal prediction supplies coverage on the keypoints, geometry carries that coverage to the pose as PURSE, and semidefinite relaxation turns the set into concrete worst-case bounds. Because the wrapper sits on top of any heatmap-based keypoint detector, it is largely detector-agnostic.

The honest boundaries are worth stating. The coverage guarantee rests on exchangeability between calibration and test data, which holds on LM-O because poses were sampled roughly uniformly but would need care in a streaming or strongly correlated setting. The worst-case bounds are correct but can be conservative, sometimes much larger than the actual error, and the paper studies ways to tighten them. And the guarantee is on the keypoints and the resulting set, not a promise that any single point estimate is within a fixed distance. Even with those caveats, the framework points toward provably correct perception for other geometric vision problems where safety, not just benchmark score, is what matters.